Div A **Python Class Assignment** **Dt:14/10/2026**

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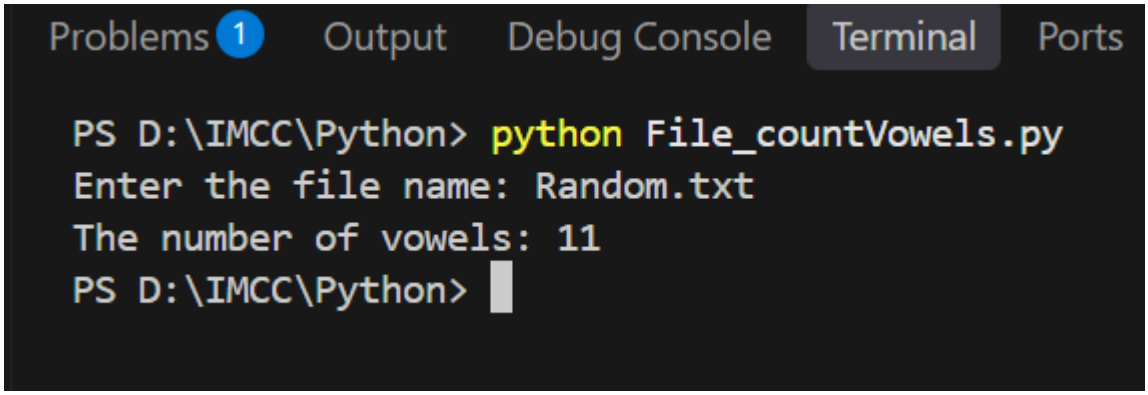
1. Write a program to read a file and count how many vowels are present in its content.

Soln:-

File_countVowels.py > ...

```
# Program to read a file and count the number of vowels in it.
name = input("Enter the file name: ")
with open(name, "r") as f:
    data = f.read()
    vowels = "aeiouAEIOU"
    count = 0
    for char in vowels:
        count += data.count(char)
    print("The number of vowels:", count)

```



2. Read all lines from a file into a list. Print the total number of lines and display the 2nd and 4th line.

Soln:-

File_Lines.py > ...

```python
# Read all lines from a file into a list.Print the total no. of lines
# and display the  2nd and 4th line.
with open("Random.txt", "r") as f:
    lines = f.readlines()
    print("Total number of lines in the file:", len(lines))
    if len(lines) >= 4:
        print("The 2nd line is:", lines[1].strip())
        print("The 4th line is:", lines[3].strip())
    else:
        print("The file does not have enough lines to display the 2nd and 4th lines.")
```

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```
PS D:\IMCC\Python> python File_Lines.py
Total number of lines in the file: 5
The 2nd line is: Using
The 4th line is: hello world!
PS D:\IMCC\Python>
```

3. Take a sentence from the user, split it into words, and store them in a list. Print the longest word.

Soln:-

Sentence.py > ...

```python
# Take  sentence from the user , split it into words and store the
sentence = input("Enter a sentence: ")
words = sentence.split()
print("The words in the sentence are:", words)
longest_word = max(words, key=len)
print("The longest word in the sentence is:", longest_word)
```

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```
PS D:\IMCC\Python> python sentence.py
Enter a sentence: Hello world! How you doing?
The words in the sentence are: ['Hello', 'world!', 'How', 'you', 'doing?']
The longest word in the sentence is: world!
PS D:\IMCC\Python>
```

4. Read a file, spilt its content into words, and store them in a list.
    i. Print the total number of words.
    ii. Print all unique numbers.

Soln:-

FileOp.py > ...

```python
# Read a file, split its content into words, and store them in a list.
# i. print the total no. of words. ii. print all unique words.
file_name = input("Enter the file name: ")
with open(file_name, "r") as f:
    content = f.read()
    words = content.split()
    print("Total number of words in the file:", len(words))
    unique_words = set(words)
    print("Unique words in the file:", unique_words)
```

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```
PS D:\IMCC\Python> python FileOp.py
Enter the file name: Random.txt
Total number of words in the file: 8
Unique words in the file: {'hello', 'Bye!', 'world!', 'Using', 'Hi!', 'There', 'opertions.', 'file'}
PS D:\IMCC\Python>
```